SMART CONTROLLER

A PROJECT REPORT

Submitted by

Vishv Patel (176940307546)

In partial fulfillment for the award of the degree of

Department of Diploma in Computer Engineering

ALPHA COLLEGE OF ENGINEERING AND TECHNOLOGY, GANDHINAGAR

Gujarat Technological University, Ahmedabad

\

CERTIFICATE

Date:

This is to certify that the Project entitled "SMART CONTROLLER" has been carried out by ***Vishv Patel (176940307546)*** *under* my guidance in partially fulfillment of the Diploma of *Department of Computer Engineering 5*th Semester of Gujarat Technological University, Ahmedabad during the academic year 2019 - 20.

Internal Guide
Mr. Meet Patel
Computer Engineering Department

Head Of Department
Dr. Dushyantsinh Rathod
Computer Engineering Department

Alpha College of Engineering and Technology, Gandhinagar

Department of Diploma Computer Engineering

2019

ACKNOWLEDGEMENT

We could have accomplished nothing without the moral support and guidance from our **organization & friends.** Many people have contributed to make this project a reality, and we would like to thank them all. We express my sincere gratitude to **Mr. Meet Patel** and for their timely help and guidance at every stage of execution and development of project. Whenever any student has problem they paid attention to him.

With Sincere Regards

Vishv Patel (176940307546)

ABSTRACT

SMART CONTROLLER

Smart homes are increasing these days. People have smartphones with them all the time. In this project Arduino Uno plays a key role by processing the data. The proposed of this project is controls the any home electrical appliances with one click or voice command using smart phones. Commands are sent via Bluetooth Technology to Arduino Uno. So, you can't need to get up and Turn Off/On Switches you can directly On/Off any home appliances by clicking one button on smartphones.

LIST OF FIGURES

Figure No	Figure Descriptions	Page No
1.4.3	Arduino Architecture	8
2.1.1	Waterfall Model	10
4.1.1.1	Class Diagram	18
4.5.1.1	Context Level Diagram	18
4.5.1.2	1st Level DFD(Admin)	19
4.5.1.3	1st Level DFD(User)	19
4.5.3.1	ER Diagram	21
4.6.1.1	Use-Case Diagram (Admin)	22
4.6.1.2	Use-Case Diagram (User)	23
4.6.2.1	Sequence Diagram	23
4.6.3.1	Activity Diagram (Admin)	24
4.6.3.2	Activity Diagram (User)	25
4.7.1	Circuit Diagram	26

LIST OF TABLES

Table No	Table Name	Page No
4.5.2.1	Admin	20
4.5.2.2	User	20
4.5.2.3	Appliances	20
4.8.1	Truth Table	26

LIST OF SYMBOLS

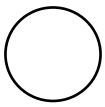
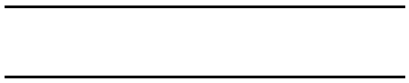
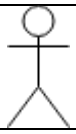
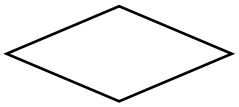
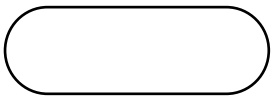
No	Symbols	Name
1		Entity
2		Process
3		Arrow
4		Data Store
5		Actors
6		Relationship
7		Start
8		Terminate
9		Action

TABLE OF CONTENT

Chapter	Content	Page No
	Abstract	1
	List of figures	2
	List of tables	2
	List of symbols	3
	Table of content	4
1	**Introduction**	**6**
1.1	Project Summary	7
1.2	Purpose	7
1.3	Scope	7
1.4	Technology Review	7
1.4.1	Introduction About Technology	7
1.4.2	Introduction About Software	7
1.4.3	Arduino Architecture	8
2	**Project Management**	**9**
2.1	Project Management	10
2.2	Risk Management	12
2.2.1	Risk Identification	12
2.2.2	Risk Analysis	12
2.2.3	Risk Planning	13
3	**System Requirement Study**	**14**
3.1	User Characteristics	15
3.2	Hardware And Software Characteristics	15
4	**System Analysis And Design**	**16**
4.1	Study Of Current System	17
4.2	Feasibility Test	17
4.3	Requirement Validation	17
4.4	Structure View	17
4.4.1	Class Diagram	17
4.5	Functional Diagram	18
4.5.1	Data Flow Diagram	18
4.5.2	Data Dictionary	19
4.5.3	Entity Relationship Diagram	20
4.6	Behavioral Modeling	21
4.6.1	Use-Case Diagram	21
4.6.2	Sequence Diagram	23
4.6.3	Activity Diagram	24
4.7	Circuit Diagram	25
4.8	Truth Table	26
5	**Enhancements**	**27**
5.1	Future Enhancements	28

6 Conclusion 29
6.1 Conclusion 30

7 Snap Shots 31

8 References 34
8.1 References 35

CHAPTER 1
INTRODUCTION

1.1 Project Summary

Smart homes are increasing these days. People have smartphones with them all the time. In this project Arduino Uno plays a key role by processing the data. The proposed of this project is controls the any home electrical appliances with one click or voice command using smart phones. Commands are sent via Bluetooth Technology to Arduino Uno. So, you can't need to get up and Turn Off/On Switches you can directly On/Off any home appliances by clicking one button on smartphones.

1.2 Purpose

The main purpose of this project is to control any home appliances with smart phones.

1.3 Scope

People use Their smartphone as a controller to control A.C, Fan, Light, T.V when they are busy in their works and some senior citizen they can't get up properly so they use this smart controller

1.4 Technology Review

1.4.1 Introduction About Technology

- C/C++, Java, Python are used for programming Arduino. It requires a software called Arduino IDE to write & upload C/C++ to Arduino.
- Arduino Uno is programmed using Arduino IDE software that you can download from arduino.cc.
- MIT App Inventor software was used to create the Android app (.apk) for this project.

1.4.2 Introduction About Software

Arduino IDE: The Arduino integrated development environment (IDE) is a cross-platform application (for Windows, macOS, Linux) that is written in the programming language Java. ... The Arduino IDE supplies a software library from the Wiring project, which provides many common input and output procedures.

App Inventor: App Inventor is a web application integrated development environment originally provided by Google, and now maintained by the Massachusetts Institute of Technology (MIT). It allows newcomers to computer programming to create application software (apps) for two operating systems (OS): Android, and iOS, which, as of 8 July 2019, is in final beta testing, scheduled to be released publicly in summer 2019. It is free and open-source software released under dual licensing: a Creative Commons Attribution ShareAlike 3.0 Unported license, and an Apache License 2.0 for the source code.

1.4.3 Arduino Architecture

\

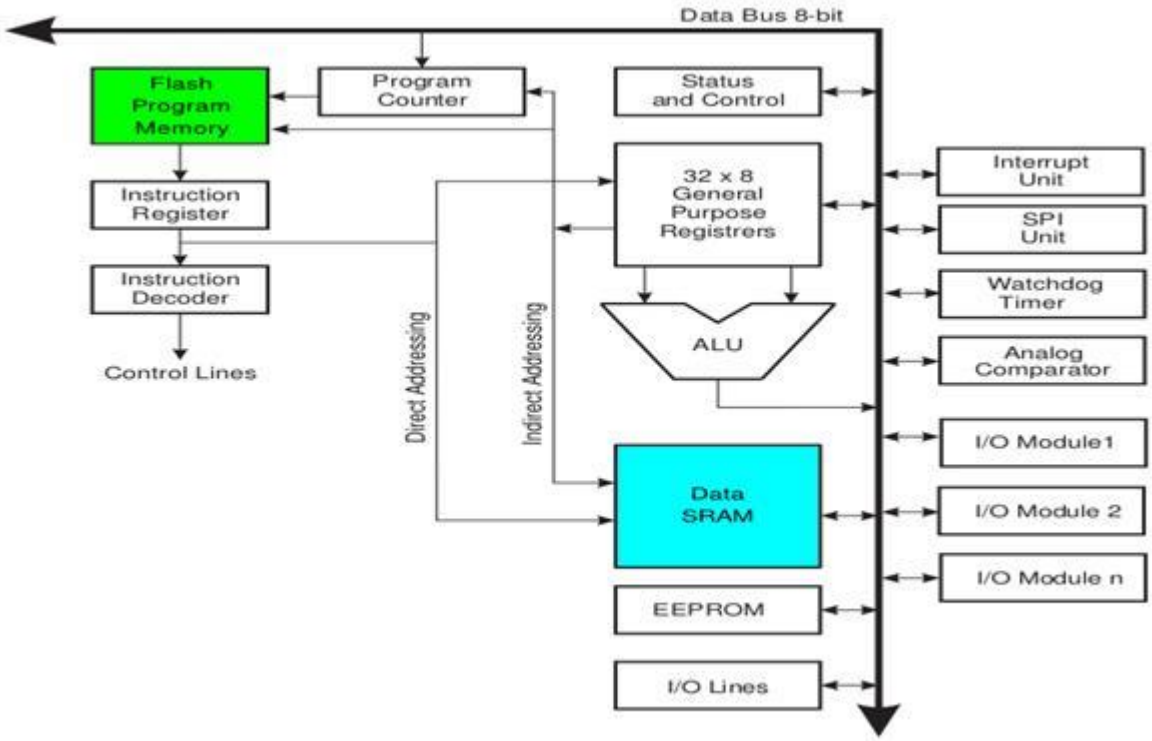


Figure 1.4.3: Arduino Architecture

Arduino Architecture

Basically, the processor of the Arduino board uses the Harvard architecture where the program code and program data have separate memory. It consists of two memories such as program memory and data memory. Wherein the data is stored in data memory and the code is stored in the flash program memory. The Atmega328 microcontroller has 32kb of flash memory, 2kb of SRAM 1kb of EPROM and operates with a 16MHz clock speed.

CHAPTER 2
PROJECT MANAGEMENT

2.1 Project Management

Waterfall Model

The Waterfall Model was the first Process Model to be introduced. It is very simple to understand and use. In a Waterfall model, each phase must be completed before the next phase can begin and there is no overlapping in the phases. Waterfall model is the earliest SDLC approach that was used for software development.

In "The Waterfall" approach, the whole process of software development is divided into separate phases. The outcome of one phase acts as an input for the next phase sequentially. This means that any phase in the development process begins only if the previous phase is complete. The waterfall model is a sequential design process in which progress is seen as flowing steadily downwards (like a waterfall) through the phases of Conception, Initiation, Analysis, Design, Construction, Testing, Production/Implementation and Maintenance.

As the Waterfall Model illustrates the software development process in a linear sequential flow; hence it is also referred to as a Linear-Sequential Life Cycle Model.

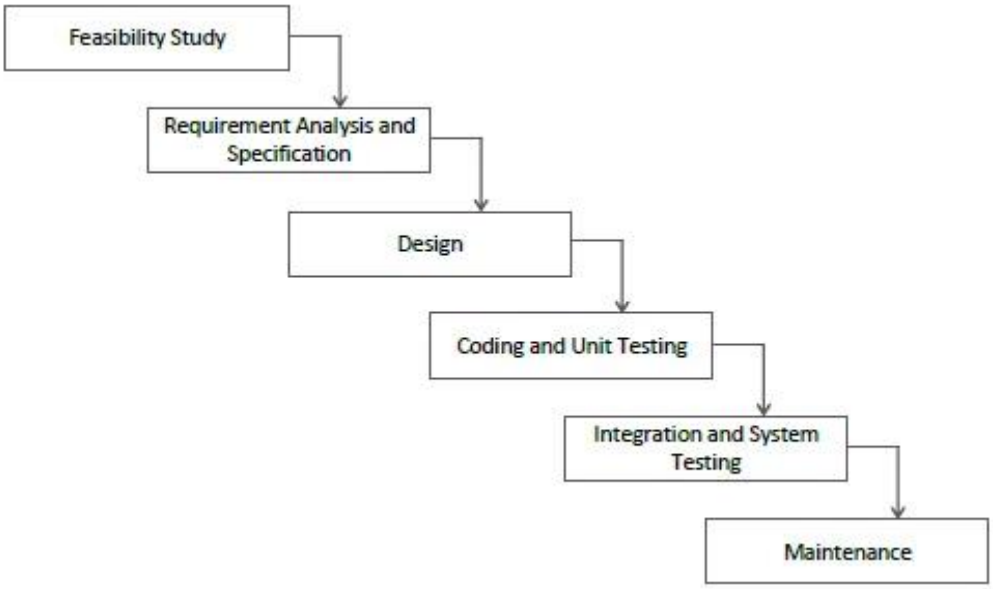


Figure 2.1.1 Waterfall Model

Sequential Phases in the Waterfall Model

- **Requirements:** The first phase involves understanding what needs to design and what is its function, purpose, etc. Here, the specifications of the input and output or the final product are studied and marked.

- **System Design:** The requirement specifications from the first phase are studied in this phase and system design is prepared. System Design helps in specifying hardware and system requirements and also helps in defining overall system architecture. The software code to be written in the next stage is created now.

- **Implementation:** With inputs from system design, the system is first developed in small programs called units, which are integrated into the next phase. Each unit is developed and tested for its functionality which is referred to as Unit Testing.

- **Integration and Testing:** All the units developed in the implementation phase are integrated into a system after testing of each unit. The software designed, needs to go through constant software testing to find out if there are any flaw or errors. Testing is done so that the client does not face any problem during the installation of the software.

- **Maintenance:** This step occurs after installation and involves making modifications to the system or an individual component to alter attributes or improve performance. These modifications arise either due to change requests initiated by the customer, or defects uncovered during live use of the system. The client is provided with regular maintenance and support for the developed software.

All these phases are cascaded to each other in which progress is seen as flowing steadily downwards (like a waterfall) through the phases. The next phase is started only after the defined set of goals are achieved for the previous phase and it is signed off, so the name “Waterfall Model “.

Advantages of the Waterfall Model

1. The advantage of waterfall development is that it allows for departmentalization and control. A schedule can be set with deadlines for each stage of development and a product can proceed through the development process model phases one by one.
2. The waterfall model progresses through easily understandable and explainable phases and thus it is easy to use.
3. It is easy to manage due to the rigidity of the model – each phase has specific deliverables and a review process.
4. In this model, phases are processed and completed one at a time and they do not overlap. Waterfall model works well for smaller projects where requirements are very well understood.

Disadvantages of Waterfall Model

1. It is difficult to estimate time and cost for each phase of the development process.
2. Once an application is in the testing stage, it is very difficult to go back and change something that was not well-thought out in the concept stage.
3. Not a good model for complex and object-oriented projects.
4. Not suitable for the projects where requirements are at a moderate to high risk of changing.

2.2 Risk Management

Risk management is the process of measuring, or assessing, risk and developing strategies to manage it. Strategies include transferring the risk to another party, avoiding the risk, reducing the negative effect of the risk, and accepting some or all of the consequences of a particular risk. Traditional risk management focuses on risks stemming from physical or legal causes (e.g. natural disasters or fires, accidents, death, and lawsuits). Financial risk management, on the other hand, focuses on risks that can be managed using traded financial instruments. In ideal risk management, a prioritization process is followed

whereby the risks with the greatest loss and the greatest probability of occurring are handled first, and risks with lower probability of occurrence and lower loss are handled later. Steps in the risk management process

Establishing the context involves: -

1. Planning the remainder of the process.
2. Mapping out the following: the scope of the exercise, the identity and objectives of stakeholders, and the basis upon which risks will be evaluated.
3. Defining a framework for the process and an agenda for identification.
4. Developing an analysis of risk involved in the process.

2.2.1 Risk Identification

After establishing the context, the next step in the process of managing risk is to identify potential risks. Risks are about events that, when triggered, cause problems. Hence, risk identification can start with the source of problems, or with the problem itself. In this project there can be following risks.

The other risk is associated with the software. If in the software the wrong user is

- Authorized by mistake then he may do changes that cause the system in dangerous mode.
- There can be risk of natural threats.

2.2.2 Risk Analysis

Once risks have been identified, they must then be assessed as to their potential severity of loss and to the probability of occurrence. Regardless of the prevention techniques employed, possible threats that could arise inside or outside the organization need to be assessed. Regardless of the type of threat, the goals of business recovery planning are to ensure the safety of customers, employees and other personnel during and following a disaster.

The relative probability of a disaster occurring should be determined. Here by the first risk can occur because of the lack of communication with all branches of Apollo for requirement fulfilment. for example, the company may not have interacted with the branch of Apollo in U.S.A. and that branch needs some additional functionality

of the software. If by mistake any person threat Administrator password, then he can change the data in software and can leak information. Same thing occurs if the wrong user is authorized.

The software may be in problem by natural threat e.g. internal flooding, external flooding, internal fire, external fire etc.

2.2.3 Risk Planning

Once risks have been identified and assessed, all techniques to manage the risk fall into one or more of these four major categories:

- Tolerate (retention)
- Treat (mitigation)
- Terminate (elimination)
- Transfer (buying insurance)

Ideal use of these strategies may not be possible. Some of them may involve trade-offs that are not acceptable to the organization or person making the risk management decisions.

CHAPTER 3
SYSTEM REQUIREMENT STUDY

3.1 User Characteristics

It describes the type of user which deals with the applications. Basically, this application has two types of users as given below:

1. Admin
2. User

Admin: Admin can add all the modules in an application

User: User can operate all those appliances using their smartphones

3.2 Hardware and Software Characteristics

Hardware requirement for Android Application Development:

- Minimum intel dual core 1.5 GHz processor
- Minimum 2GB of RAM
- 5 GB free space in Hard Disk storage

Requirements to run the App:

- Compatible to all Android devices above 4.0
- 2 GB RAM
- 5 MB internal free storage to install app
- Working without internet connection.

CHAPTER 4
SYSTEM ANALYSIS AND DESIGN

4.1 Study of Current System

In a current day we have to do manually switch on/off our home appliances. So that's why we make this system. In this system we can operate by using smartphone or voice command. we can also operate manually

4.2 Feasibility Study

An important outcome of the preliminary investigation is the determination that the system is feasible or not. The main aim of the feasibility study activity is to determine whether it would be financially and technically feasible to develop a project.

Operational Feasibility: The system contributes to the overall objective of the organization because it meets all the requirements given by the organization. We have tested our system under different test face so there is no chance for the organization that the system will suffer from undesirable problems. The system is quite easy to use and learn. User requires no special training for operating system.

Technical Feasibility: It can be easily implemented using the current technology because the system would be developed in Android because it is an open source software stack. So, a cost constraint is not a major issue. About Schedule constraints, yes, it is not going to be strictly followed due to the changing needs of the customers.

Economical Feasibility: The system can be integrated with other system which is already in place. It is cost effective because resources and the tools that are used in this system are highly optimized to get the best results and mainly for free.

4.3 Requirements Validation

- Requirement validation examines this specification to ensure that all the system requirements have been stated unambiguously.
- These inconsistent, errors have been detected and corrected and the work products confirmed to the standard.
- Source of the requirement are identified; final Statement of requirement has been examined by original source.
- Requirements related to main requirements are founds.
- Requirements are clarifying stated and are not misinterpreted.
- All sources of requirements are covered to get a maximum requirement.
- All method of finding requirements is applied.

4.4 Structure View

4.4.1 Class Diagram

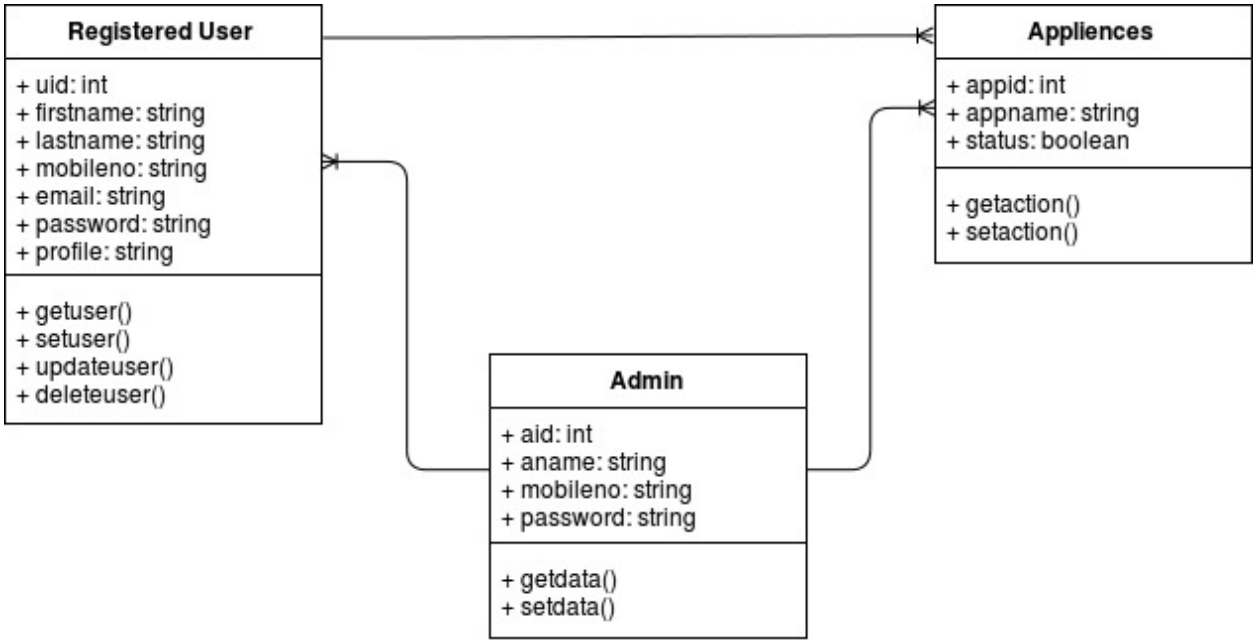


Figure 4.4.1.1: Class Diagram

4.5 Functional Modeling

4.5.1 Data Flow Diagram

Context Level DFD

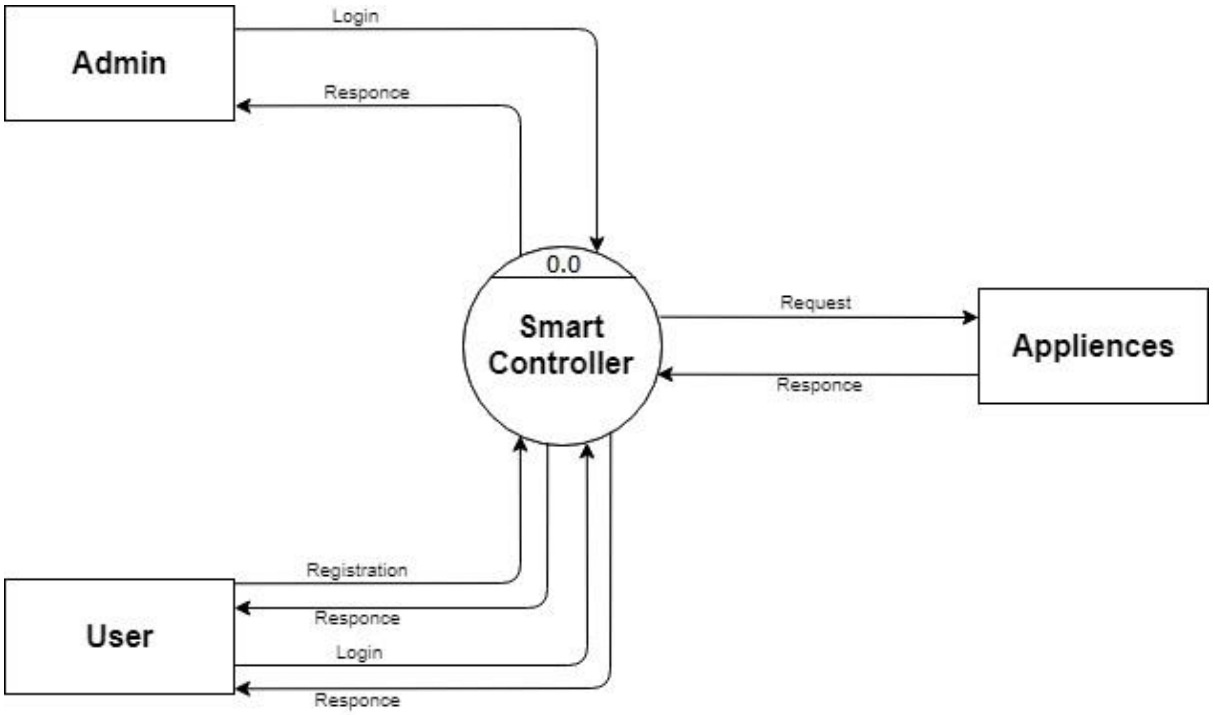


Figure 4.5.1.1: Context Level 0 DFD

1st level DFD for Admin

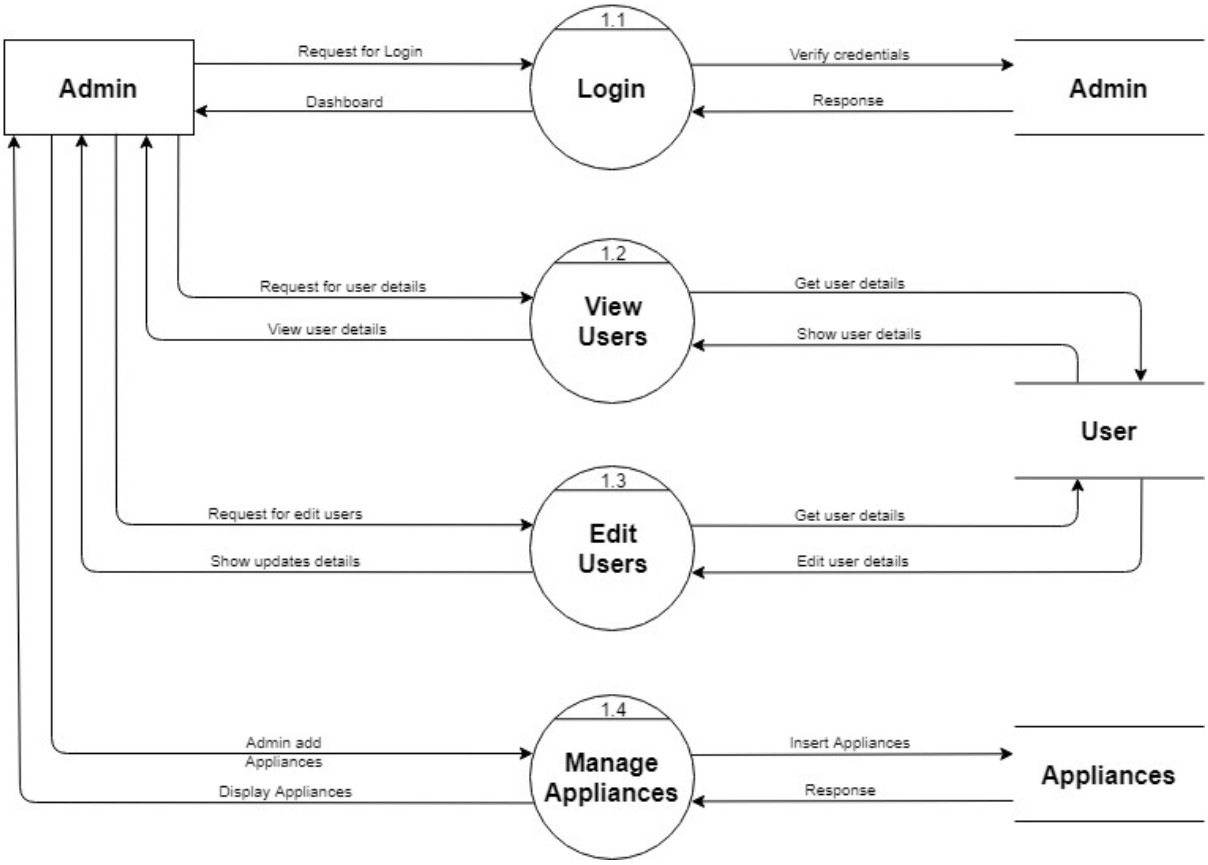


Figure 4.5.1.2 : 1st Level DFD Admin

1st level DFD for User

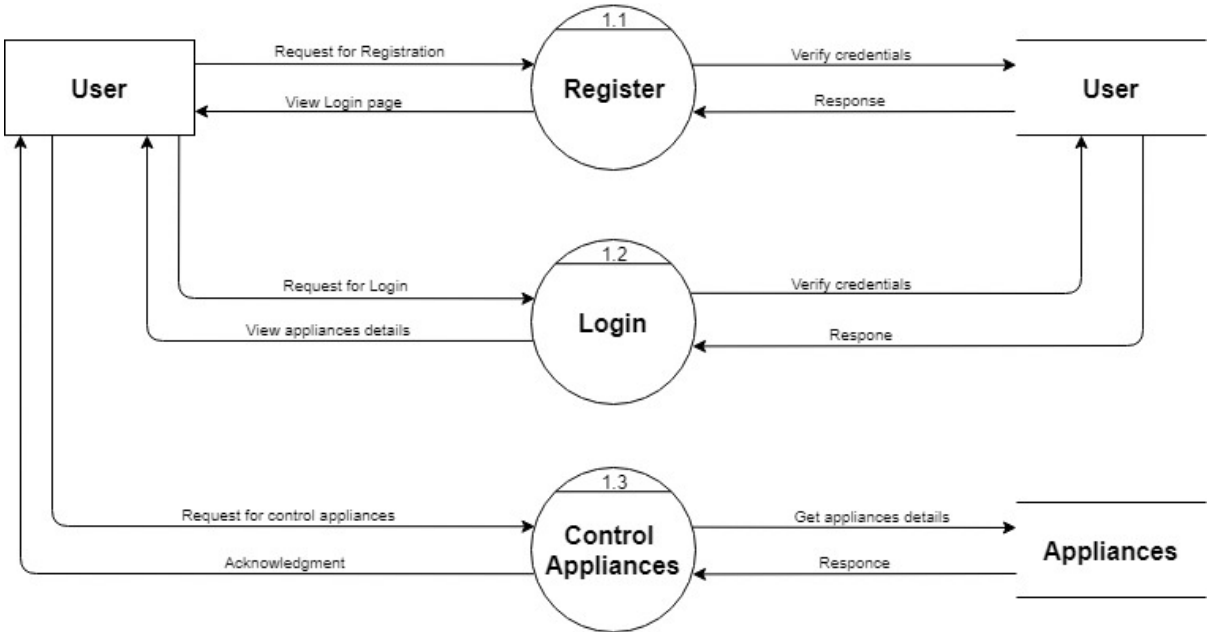


Figure 4.5.1.3: 1st Level DFD User

4.5.2 Data Dictionary

Field Name	Constraints	Datatype	Size	Description
aid	Primary Key	Number	10	Id of the admin
username	Not Null	Char	100	username of admin
password	Not Null	Varchar	255	password of admin
mobileno	Not Null	Number	10	contact number of admir

Table 4.5.2.1 : Admin

Field Name	Constraints	Datatype	Size	Description
uid	Primary Key	Number	10	User id of the user
firstname	Not Null	Char	100	First Name of the user
lastname	Not Null	Char	100	Last Name of the user
email	Not Null	Varchar	255	Email id of the user
monileno	Not Null	Number	10	Contact Number of user
password	Not Null	Varchar	255	Password of user

Table 4.5.2.2: User

Field Name	Constraints	Datatype	Size	Description
appid	Primary Key	Number	10	Id of Appliances
appname	Not Null	Char	100	Name of Appliances
status	Not Null	Boolean	-	On/Off Status

Table 4.5.2.3: Appliances

4.5.3 Entity Relationship Diagram

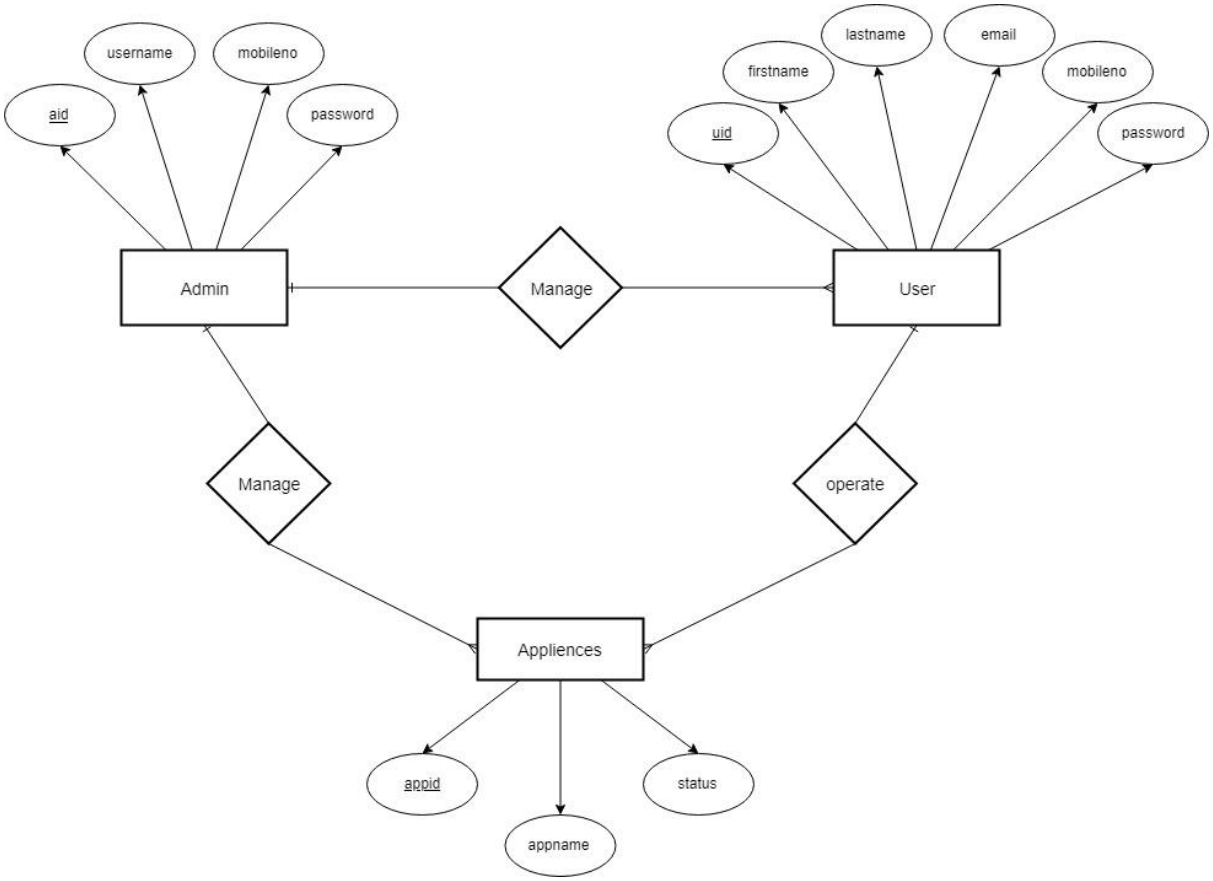


Figure 4.5.3.1: ER Diagram

4.6 Behavioral Modeling

4.6.1 Use Case Diagram

Use-Case for Admin

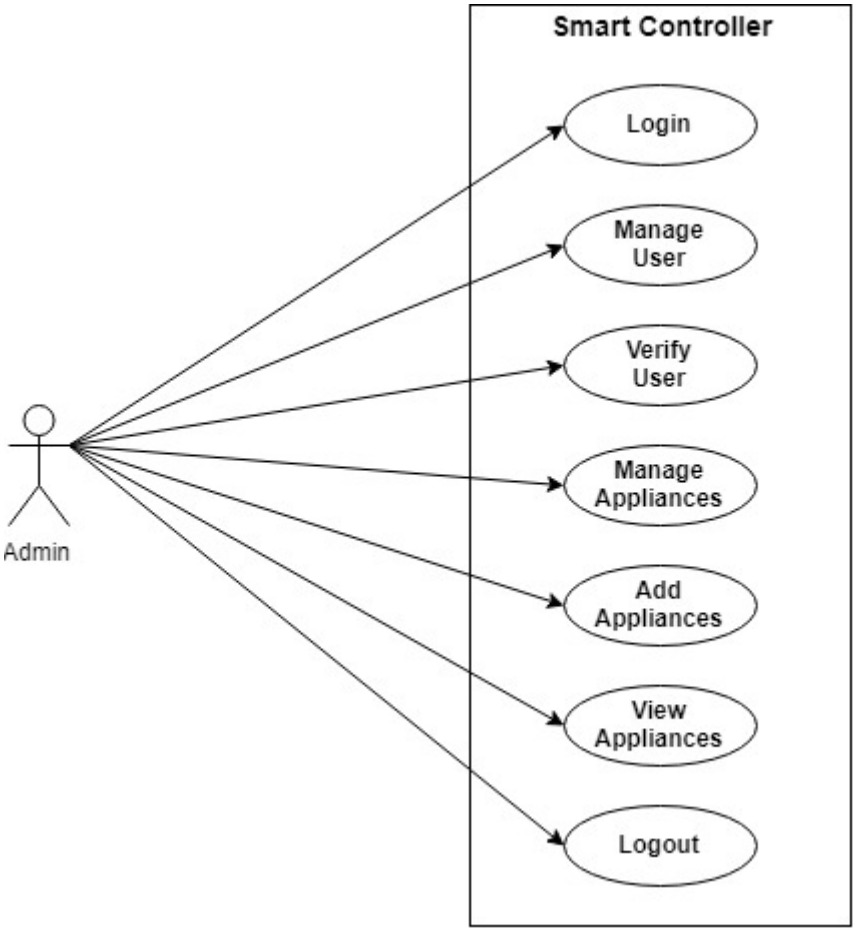


Figure 4.6.1.1: Use-Case Diagram for Admin

Use-Case for User

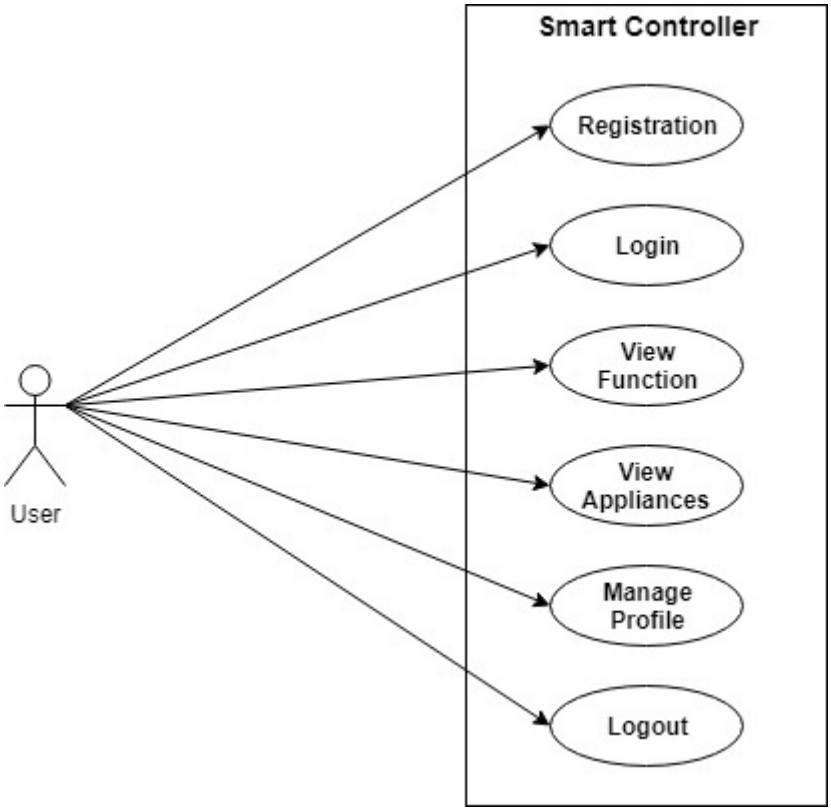


Figure 4.6.1.2: Use-Case Diagram for User

4.6.2 Sequence Diagram

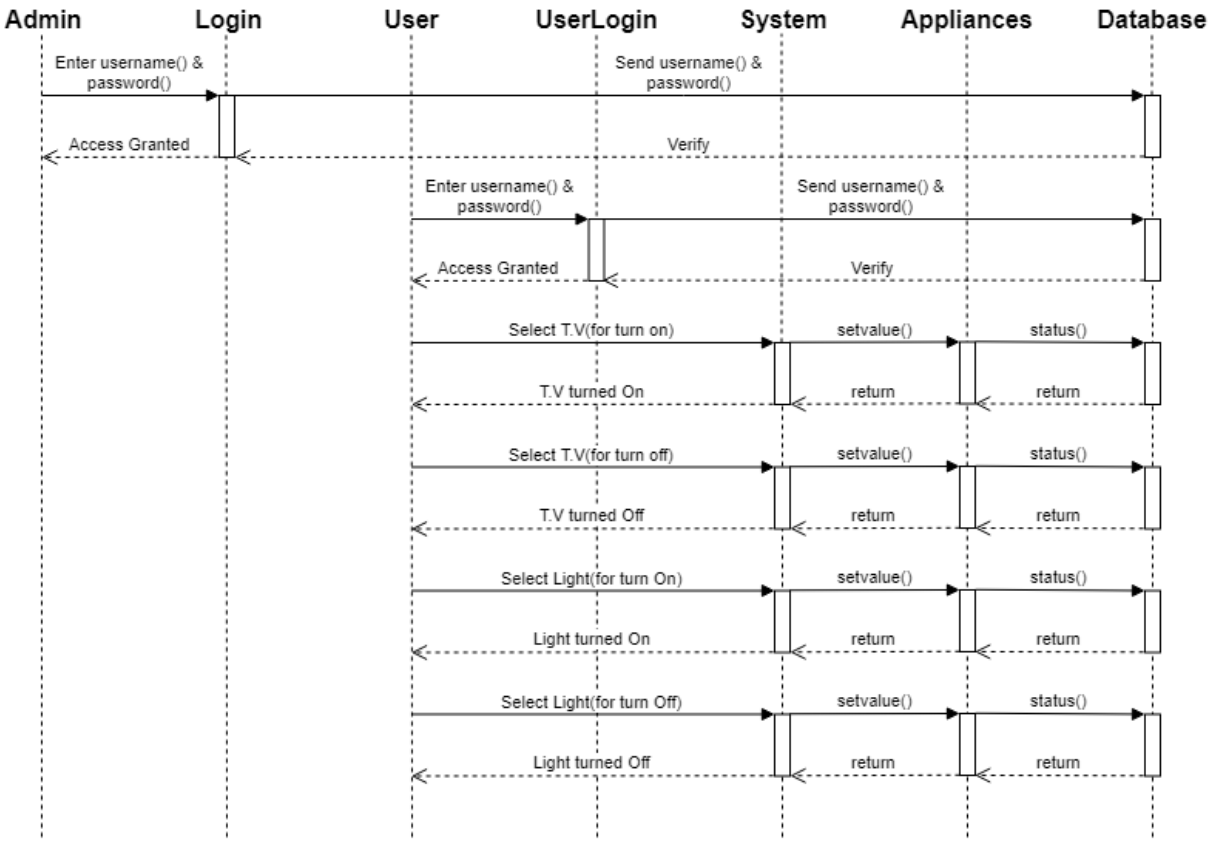


Figure 4.6.2.1: Sequence Diagram

4.6.3. Activity Diagram

Admin's Activity

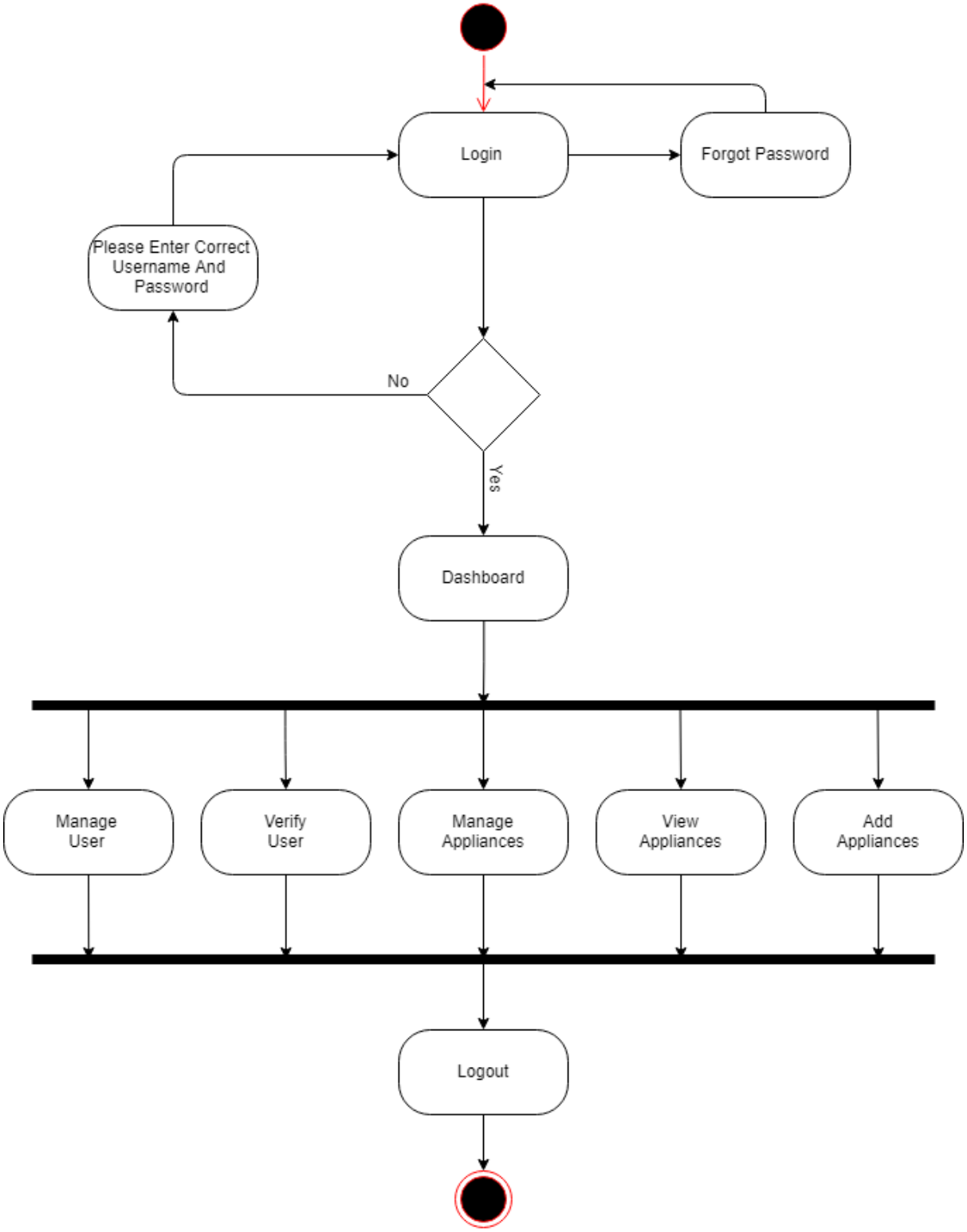


Figure 4.6.3.1: Activity Diagram for Admin

User's Activity

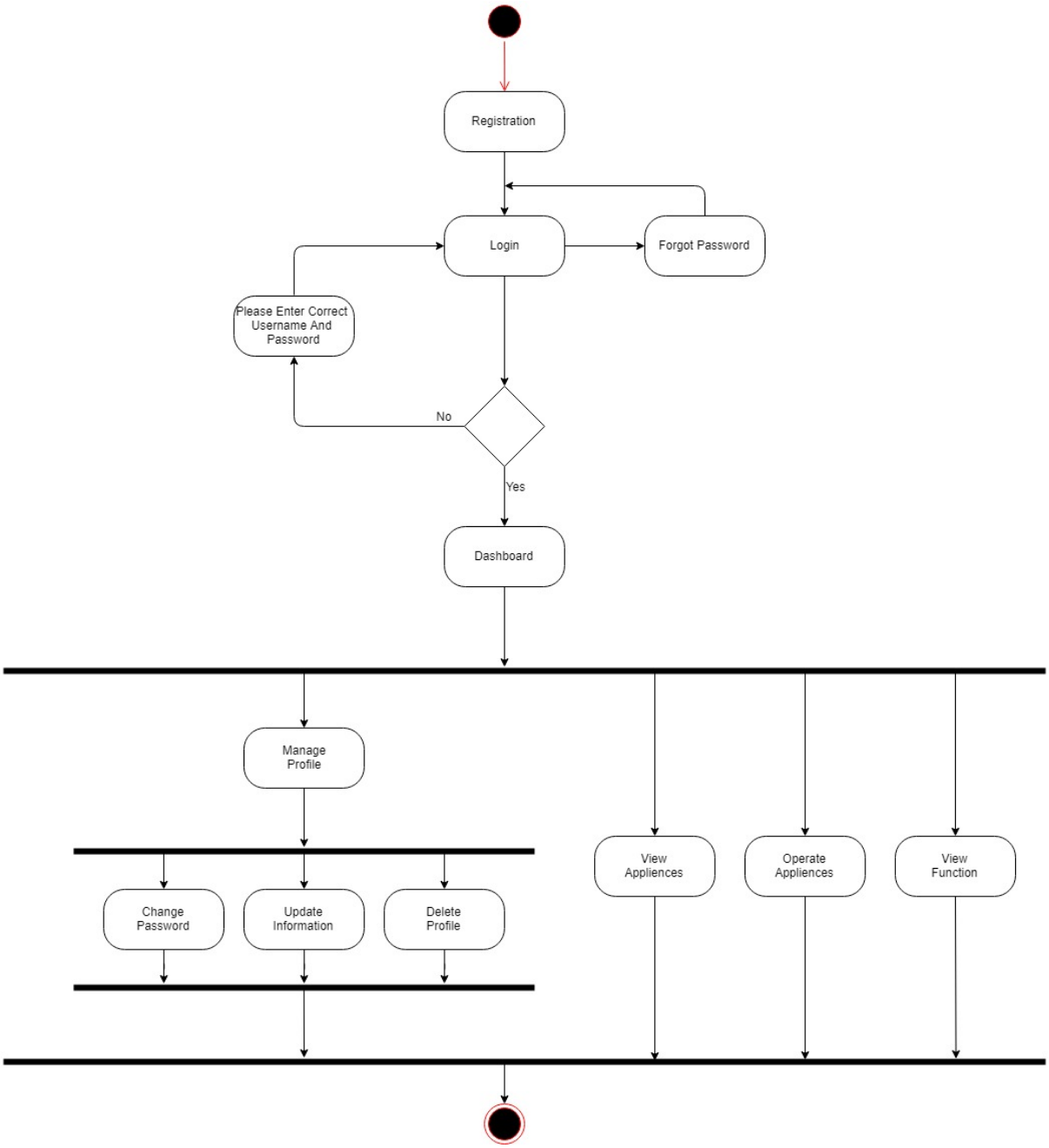


Figure 4.6.3.2: Activity Diagram for Customer

4.7 Circuit Diagram

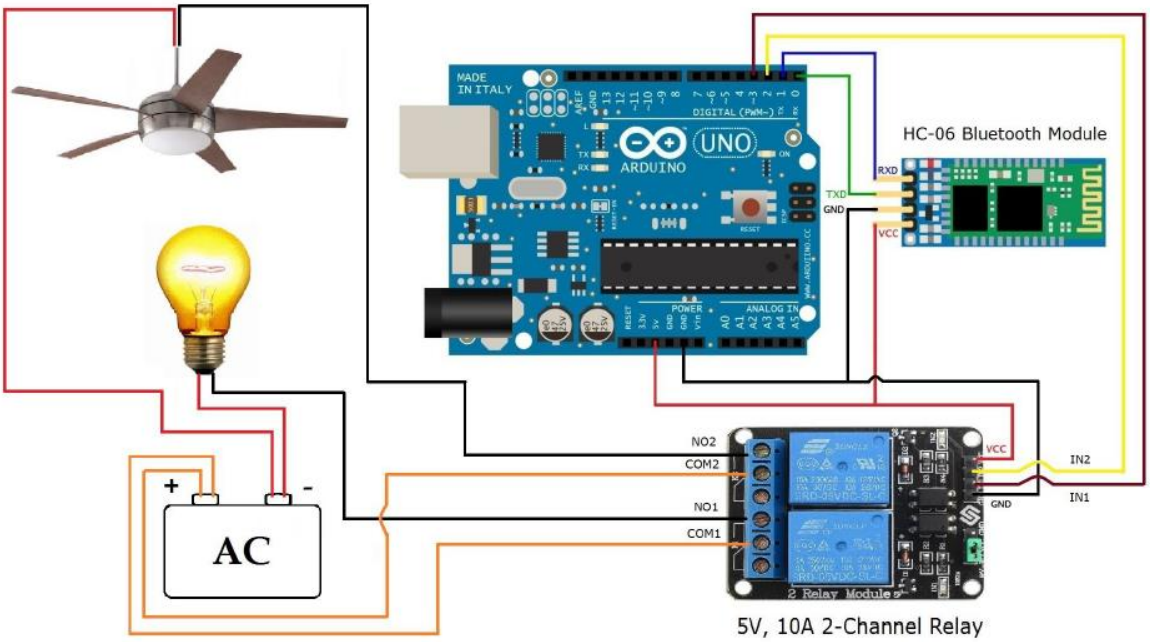


Figure 4.7.1: Circuit Diagram

4.8 Truth Table

appid	appname	status
01	Light	1
02	T.V	1
03	Fan	1
04	A.C	0

Figure 4.8.1: Truth Table

Here, status 1 means Appliances is on and 0 means its off.

CHAPTER 5
ENHANCEMENTS

5.1. Future Enhancement

In Future Enhancement we can try to make a system to IOT (Internet of Things). So, you can control all your home appliances or office appliances whenever you want and whenever you are

- We change Bluetooth module to Wi-Fi module

CHAPTER 6
CONCLUSION

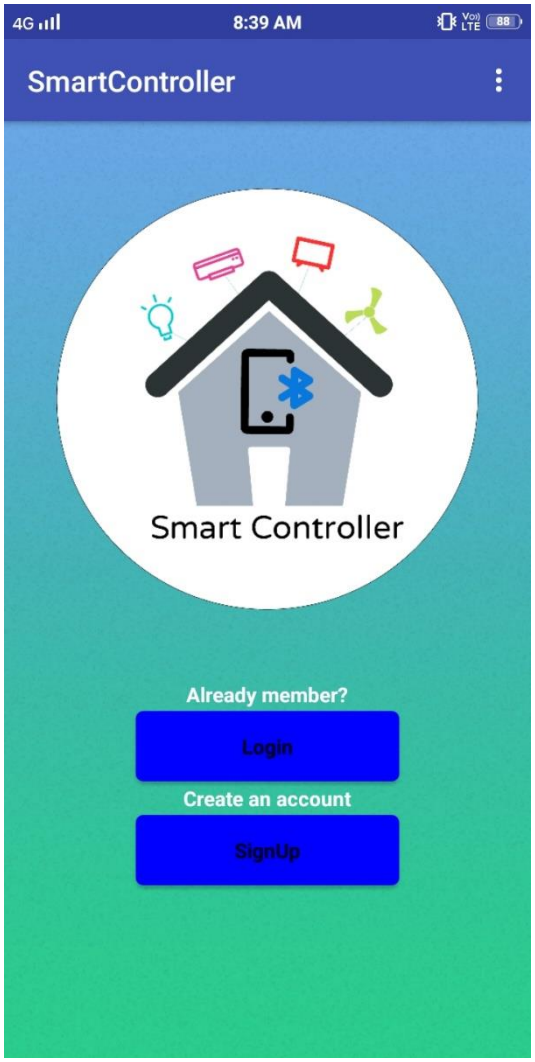
6.1. Conclusion

The home automation system has been experimentally proven to work satisfactorily by connecting sample appliances to it and appliances were successfully controlled from a wireless mobile device.

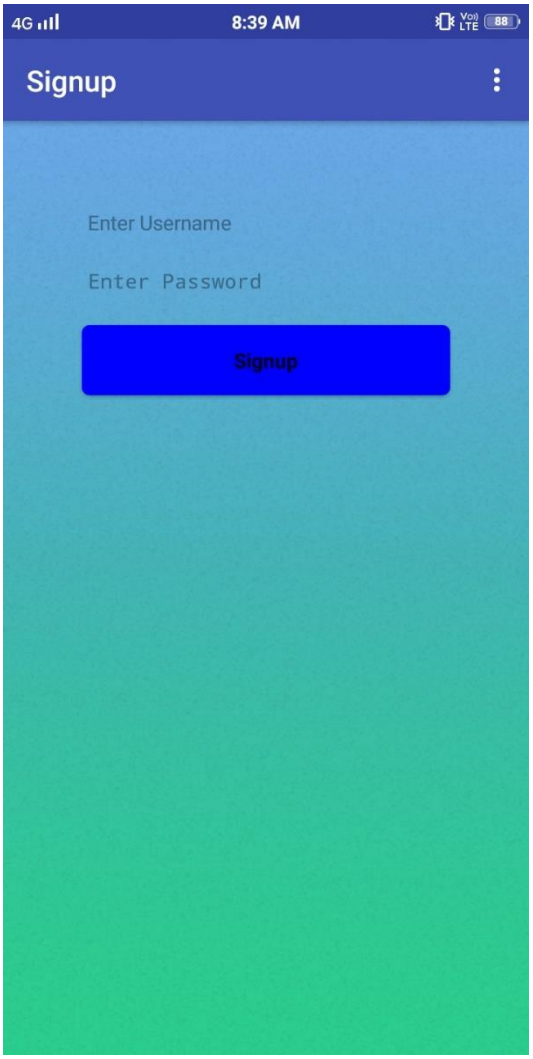
We learned many skills such as soldering, wiring the circuit and other tools that we use for this project.

The Bluetooth client was successfully tested on a multitude of different mobile phones from different manufacturers, thus proving its portability and wide compatibility. Thus, a low-cost home automation system was successfully designed, implemented and tested.

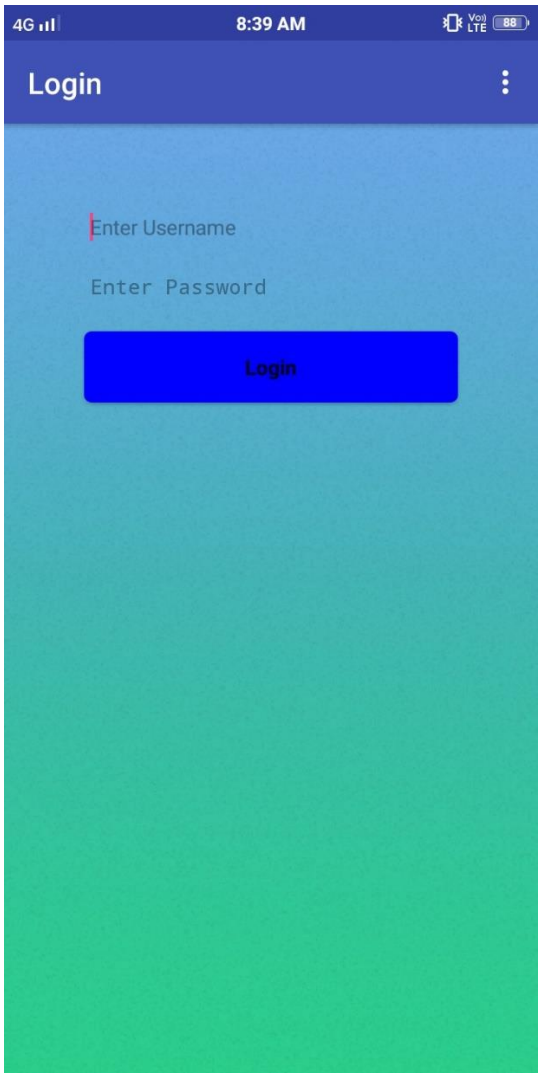
CHAPTER 7
SNAP SHOTS



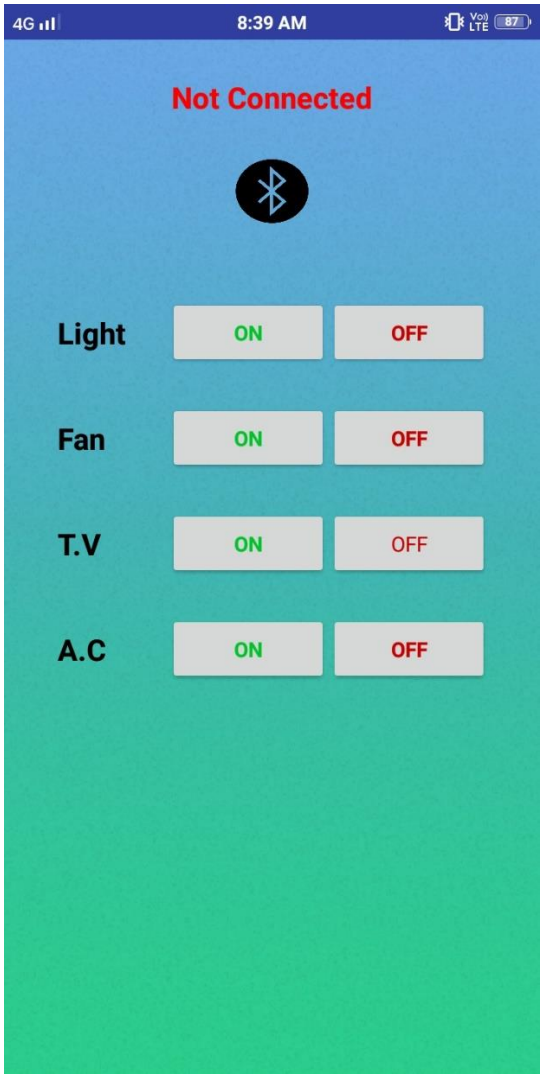
7.1 Home Screen



7.2 Sign UP Screen



7.3 Login Screen



7.4 Main Screen

CHAPTER 8
REFERENCES

8.1 References

https://www.edgefx.in/arduino-uno-board-tutorial-and-its-applications/

http://ai2.appinventor.mit.edu

https://www.arduino.cc/en/main/software

https://en.wikipedia.org/wiki/Arduino_Uno

https://www.circuito.io

https://www.draw.io/